DADS 5250 · MODULE 7 HACKATHON · TOPIC BRIEF

Rebalancing Bluebikes Across Greater Boston

Planning bike rebalancing crews by city using real Bluebikes trip data.

The problem	Decide how many full-time and seasonal rebalancing crews each city needs in each month of 2026.
Core difficulty	Bikes pile up at some stations and drain from others, demand swings with season, weather, and events, and crews cost money and take time to hire.
Data	Real Bluebikes trips, January 2024 to September 2026, plus the station list. Public, free to download. REAL
GenAI angle	Use an LLM to turn event text into structured data for your forecast, and optionally build an operations assistant (see Section 5).
Due	Sunday, October 25, 11:59 PM ET

1 Background

Bluebikes is the bike-share system for Boston and neighboring cities such as Cambridge, Somerville, and Brookline. Riders have taken over 31 million trips since it launched in 2011. In July 2026 alone, riders took 530,297 trips from 606 stations, about 17,000 a day.

Riders don't return bikes evenly. In July 2026, the station at Ames St at Main St in Kendall Square received 1,230 more bikes than left it, while Government Center lost 480 and MIT Vassar St lost 415. When a station runs empty, riders can't find a bike. When it fills up, riders can't return one. Either way, trips and revenue are lost.

Rebalancing crews drive vans that move bikes from full stations to empty ones. Demand changes a lot with the season, the weather, the academic calendar, and events like Red Sox games, the Boston Marathon, and September move-in.

Your goal is to build a month-by-month plan for 2026 that decides how many rebalancing crews each city needs, using real trip data from 2024 and 2025. You can then check your plan against what actually happened from January to September 2026.

2 Key Rules & Workflow

The trip and station data is REAL. Crew capacity, costs, and starting crew levels are planning assumptions ASSUMED given for this challenge, the same way a business case gives salaries. Use them as stated.

Stations & demand REAL

- Each station has a fixed number of docks, listed in the station file.
- Each station belongs to a city (the `Municipality` column in the station file). Plan crews by city.

- The trip data shows every pickup and return, so you can measure how many bikes each station gains or loses by hour and by day.

Measuring rebalancing need

- The data doesn't record empty or full stations, or lost trips. Your team must estimate them.
- One simple approach: assume each station starts the day half full, track bikes in and out hour by hour, and count the bikes that would need to be moved to keep it between empty and full. State whatever method you use.

Crews ASSUMED

- A crew is one driver and one van. Each crew can move up to 120 bikes per shift and works 6 shifts a week.
- A crew works only inside its own city.
- Crews in January 2026: Boston 6, Cambridge 4, Somerville 1, Brookline 1. All other cities share 2 crews.

Two ways to staff a crew ASSUMED

Crew type	Cost	Rules
Full-time crew	$54,000 per year for the driver, plus $1,500 per month for the van	A new hire needs 1 month to recruit and train before the first shift. Letting a driver go requires 1 month of notice and costs 25% of the driver's annual salary.
Seasonal contract crew	$7,800 per month, all-inclusive	Starts the month after you sign, with a minimum contract of 3 months and no cost to end it. At most 8 seasonal crews can work across the whole system in any month.

Revenue impact of a lost trip ASSUMED

Each ride brings in $3.25 on average. When a rider can't find a bike or a free dock, one of three things happens:

What happens	Probability	Revenue lost
The rider walks to a nearby station and still rides	40%	$0
The ride is lost	45%	$3.25
The ride is lost and the member does not renew their membership	15%	$3.25 + $25 of future revenue

3 The Challenge: A Month-by-Month Crew Plan for 2026

Decide how many full-time and seasonal crews each city needs for each month of 2026, balancing:

- **Cost efficiency:** avoid paying for idle crews in the slow winter months.
- **Rider service:** keep stations stocked and reduce lost trips and lost members.
- **Flexibility:** use full-time and seasonal crews wisely, given the hiring, release, and contract rules.

Recommended deliverables

- **Demand & imbalance model:** use 2024 and 2025 trips to estimate how many bikes need moving, by city and month.
- **Event awareness with GenAI:** collect real event information as text, such as Red Sox home games, the Boston Marathon, Head of the Charles, and college move-in dates. Use an LLM to extract the date, location, and expected crowd size, then add those to your model.
- **Dynamic staffing model:** a table showing full-time and seasonal crews by city for each month of 2026, including when you hire, release, or sign contracts.
- **Backtest:** compare your plan with the real January–September 2026 trips. Where would your crews have fallen short?
- **Analysis & justification:** a short report explaining your key decisions and assumptions, and their impact on cost and revenue.
- **Sensitivity analysis:** how your plan holds up if ridership is 20% higher or lower, or if e-bikes keep growing as a share of trips.
- **Operations assistant (optional):** a chat assistant that lets the operations team ask questions about the plan, such as "What happens if we add one crew to Cambridge in September?"
- **Other considerations:** any other factor you recommend considering, such as adding docks at busy stations or offering riders rewards to return bikes to empty stations.

4 Data Provided

All data is real and public, published by Bluebikes under the Bluebikes Data License Agreement. See the Bluebikes System Data page (https://bluebikes.com/system-data).

File	Where to get it	What it contains
Monthly trip files, January 2024 to September 2026	Bluebikes trip data bucket (https://s3.amazonaws.com/hubway-data/index.html), for example `202407-bluebikes-tripdata.zip`	One row per trip. About 6–25 MB zipped per month
Station list	The CSV link on the System Data page	Every station's number, name, latitude and longitude, city, and total docks

Each trip file has 13 columns:

Column(s)	Meaning
`ride_id`	The trip ID
`rideable_type`	`classic_bike` or `electric_bike`
`started_at`, `ended_at`	Start and end time
`start_station_name`, `start_station_id`, `end_station_name`, `end_station_id`	Where the trip started and ended
`start_lat`, `start_lng`, `end_lat`, `end_lng`	Start and end coordinates
`member_casual`	`member` or `casual`

Things to watch for in the real data

- Files are big. July 2026 alone has 530,297 rows and is about 127 MB unzipped. Aggregate to station-hour or station-day early.
- File formats vary. Most months are `.zip`, November 2025 is `.csv.zip`, and September 2026 is a plain `.csv`.
- Trips by Bluebikes staff and trips under 60 seconds are already removed.
- Station names in trip files may not exactly match the station list, so join on station ID where you can.
- Weather and event data are not included. You may add public sources, such as daily weather for Boston, and cite them.

5 Making It a GenAI Application

The hackathon asks you to build a working AI application that combines **at least two course techniques** (API calls, prompt engineering, structured output, LangChain, RAG, or agents). The planning model above is the problem; your AI application is how you solve it and put it in people's hands. The directions below are examples, not requirements. Pick one, combine several, or propose your own.

WORKFLOW AUTOMATOR

Event-Aware Demand Pipeline *(the core GenAI deliverable)*

Collect event listings and weather as text, have an LLM extract each event's date, place, nearest stations, and crowd size as structured data, and feed it into your demand forecast and crew plan.

Techniques: Structured output, prompt engineering, LangChain or agents

AI ASSISTANT

Operations Assistant

A chat assistant for dispatchers: "Which Cambridge stations will run empty before tomorrow's Red Sox game?" or "What if we add one crew to Cambridge in September?" It calls your model as a tool.

Techniques: Agents (tool calling), RAG, structured output

CONTENT PIPELINE

Weekly Crew Briefing

Turns the forecast, upcoming events, and the crew plan into a short weekly briefing for each city: expected hot spots, crews on duty, and contract decisions coming up.

Techniques: LangChain chains, prompt engineering

DATA ANALYZER

Ask Bluebikes

Ask questions about the trip data in plain English ("Show net bike flow by station in July") and get charts and short insights back.

Techniques: Agents (code or tool use), structured output

6 Scoping for One Week

- Prototype with one city (Cambridge or Boston) and a few months of trips, then scale up.
- Aggregate trips to station-hour or station-day as your first step. Don't work with raw trips in your model.
- Compare a summer month with a winter month early. The difference drives your seasonal crew decisions.
- The best hackathon projects are not the most complex. They are the most focused. A small model that works end to end beats a big one that doesn't run.

7 What to Submit

Due **Sunday, October 25, 11:59 PM ET** on Canvas (7 — Hackathon Project). This is a group assignment: one team member submits, and all members contribute to every deliverable.

- Your completed Jupyter notebook (`.ipynb`) with every cell run and all outputs visible.
- Your GitHub repo link, containing the code and a README with setup instructions, an architecture diagram, and each team member's contributions.
- A link to your 3-minute demo video walking through your product, your architecture, and the part you're most proud of.
- Review the rubric attached to the assignment to make sure your submission meets every requirement.

AI use. You may use AI tools, but explain how you used them and clearly mark AI-generated sections. You are responsible for making sure everything you submit is accurate and follows Northeastern's Academic Integrity and Use of AI policy (linked in the syllabus).

8 Choosing This Topic

Each team works on one topic. Topics are assigned **first come, first served** through the topic sign-up sheet, and each topic has a limited number of team slots. Check that this topic still has open slots, then register your team. Don't start building until your topic is confirmed on the sheet.

Questions about the topic? Post them in the Discussions Q&A Forum on Canvas so other teams can benefit, or bring them to office hours.